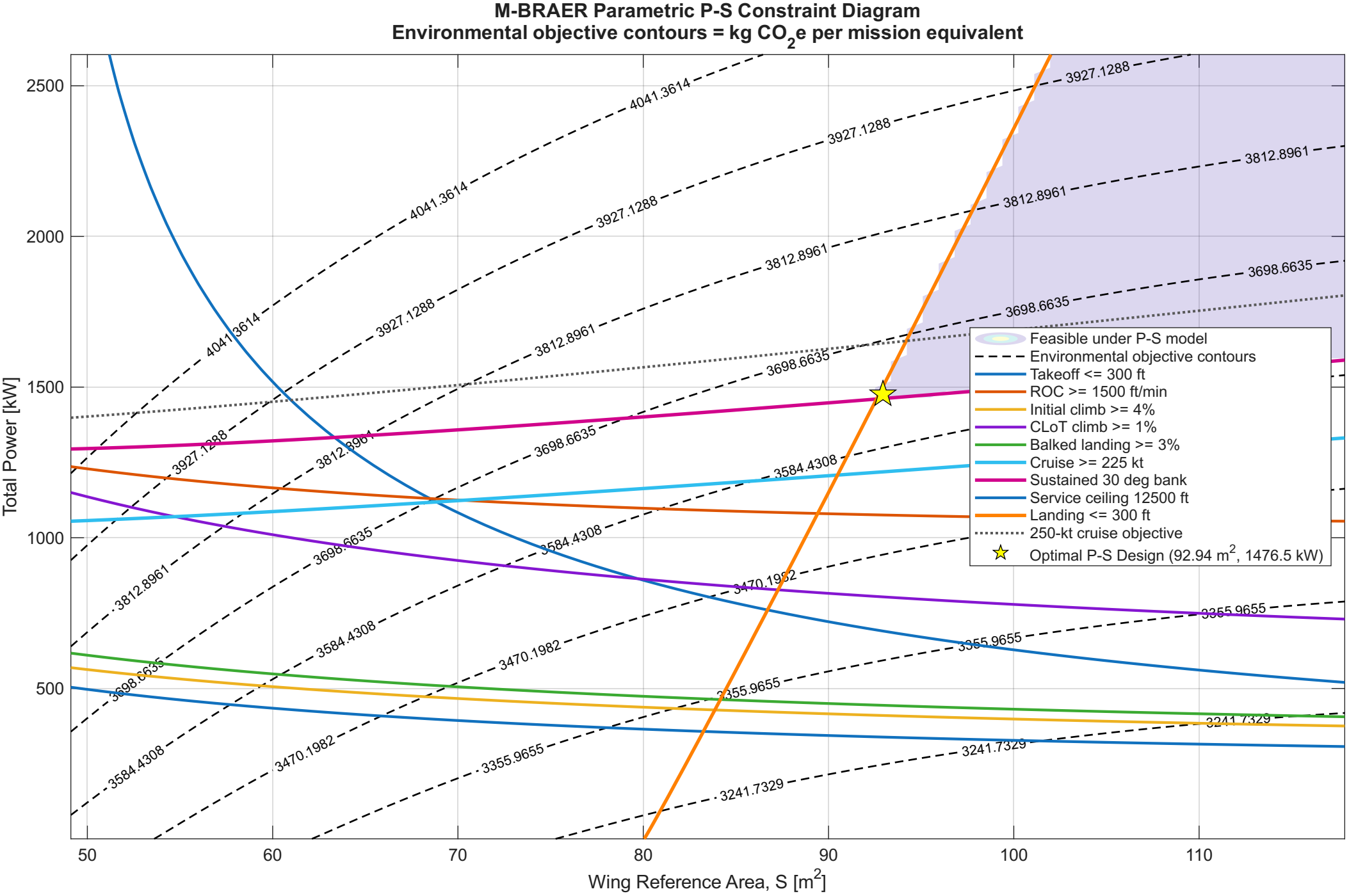
M-BRAER Parametric P-S Constraint Diagram
Environmental objective contours = kg CO2e per mission equivalent
Total Power [kW]
Wing Reference Area, S [m2]
2500
2000
1500
1000
500
50
60
70
80
90
100
110
4041.3614
3927.1288
3812.8961
3698.6635
3584.4308
3470.1982
3355.9655
3241.7329
Feasible under P-S model
Environmental objective contours
Takeoff <= 300 ft
ROC >= 1500 ft/min
Initial climb >= 4%
CLoT climb >= 1%
Balked landing >= 3%
Cruise >= 225 kt
Sustained 30 deg bank
Service ceiling 12500 ft
Landing <= 300 ft
250-kt cruise objective
Optimal P-S Design (92.94 m2, 1476.5 kW)